PROGRESS REPORT

Spontaneous Adsorption of β-Lactoglobulin (1BEB)

at the Air–Water Interface

Molecular Dynamics Simulation Study

Force Field: CHARMM36m | Water: TIP3P | GROMACS 2020.4 | April 2026

Objective

Why spontaneous? Why atomistic?

The Gap

Our previous MD studies **place the protein directly at the interface**

→ This creates a biased setup that cannot reveal:

- True adsorption pathway
- Real thermodynamic driving force
- Preferred orientation selection
- Pre-adsorption conformational changes

This Work

Fully unbiased spontaneous adsorption

→ Protein placed in bulk water center

→ No orientation bias, no pulling force

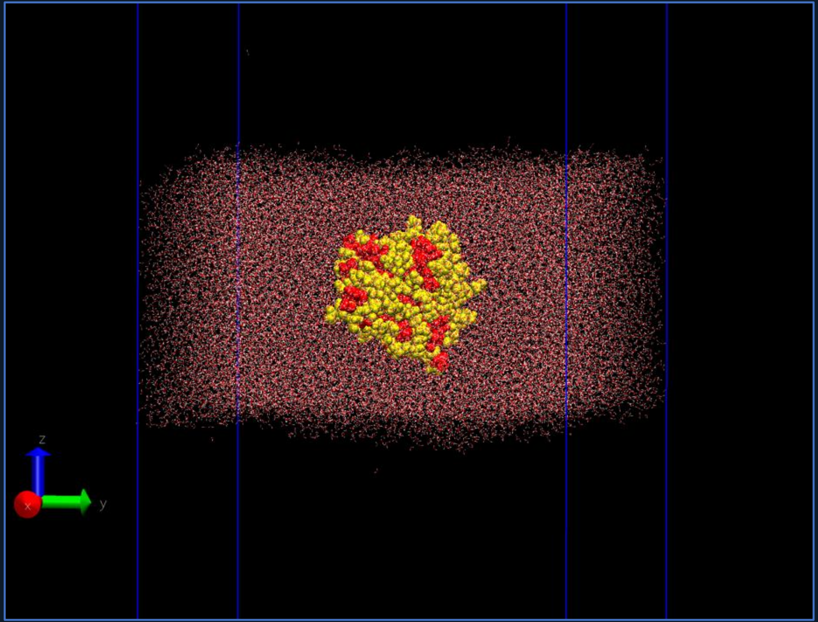
→ 1,000 ns production run

→ Atomistic CHARMM36m force field

First atomistic evidence of BLG spontaneous adsorption mechanism

System Configuration

Slab geometry with vacuum runways



Parameter	Value
Protein	β-Lactoglobulin monomer (PDB: 1BEB)
Force Field	CHARMM36m
Water	TIP3P (~55,000 molecules)
Box (Slab)	12 × 12 × 35 nm
Water Slab	~7 nm thick (CoM at Z ~17.5 nm)
Vacuum	~14 nm each side
Temperature	298 K (V-rescale)
Timestep	2 fs | VdW: force-switch 1.0–1.2 nm
pcoupl	OFF — required for slab geometry
DispCorr	OFF — CHARMM36 slab standard

Simulation Workflow

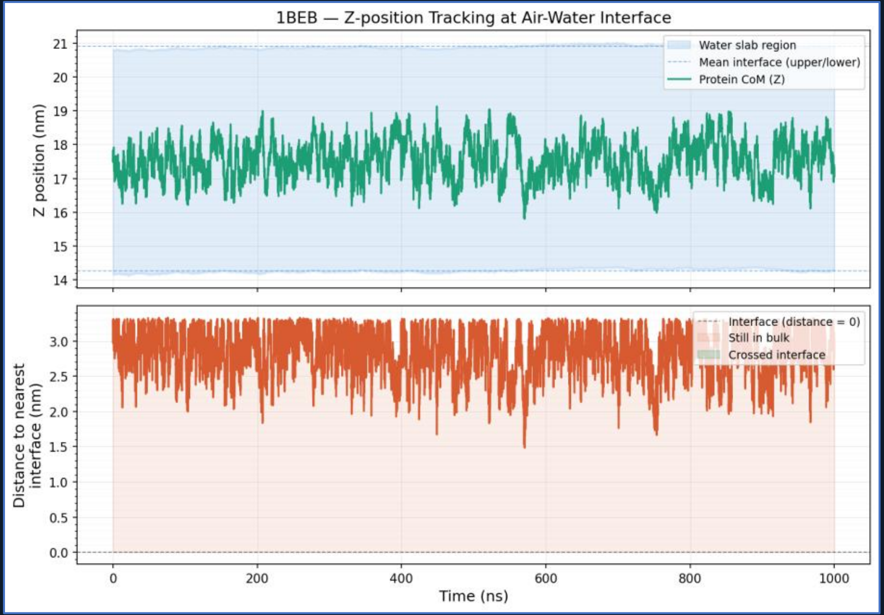
From bulk equilibration to 1 µs production

1	2	3	4	5	6
Energy Min.	NVT 100 ps	NPT 10 ns	Box Stretch	NVT Slab 10 ns	Production 1,000 ns
Steepest descent cubic box ~7 nm	298 K equilibration POSRES on	Density → ~1016 kg/m³ Parrinello-Rahman	Z: 7 → 35 nm gmx editconf	Interface equilibration POSRES on, gen_vel=yes	POSRES off Fully unbiased MD

✓ Step 6 COMPLETE — 1,000 ns finished | Performance: 104.7 ns/day | Temperature stable: 297.99 K

Diffusion Kinetics — Z-position Tracking

1,000 ns | Protein CoM vs. interface distance



3.303 nm
Distance to interface (range)

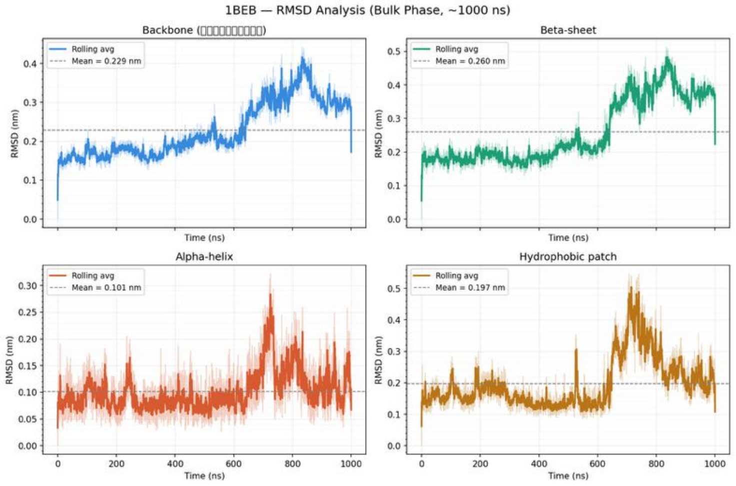
>1 µs
Adsorption timescale

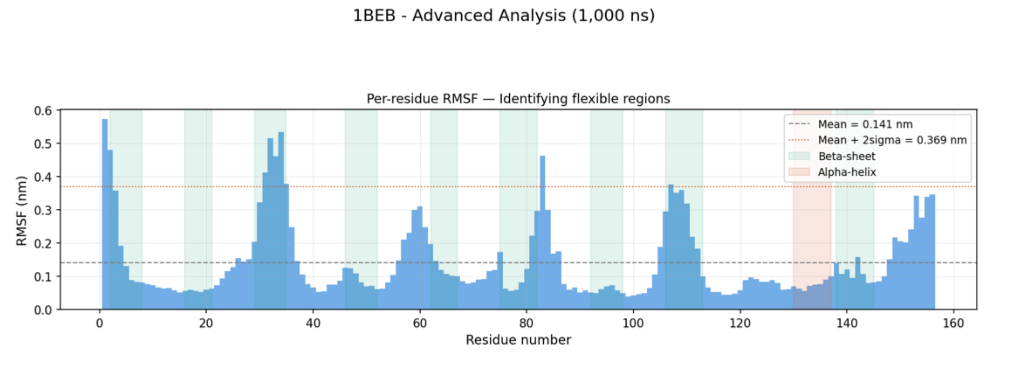
NOT YET
Adsorption status

Key Insight: BLG spontaneous adsorption from bulk center requires >1 µs — first atomistic quantification of this timescale.

Structural Analysis — RMSD & Per-residue RMSF

Stability in bulk + identifying flexible regions

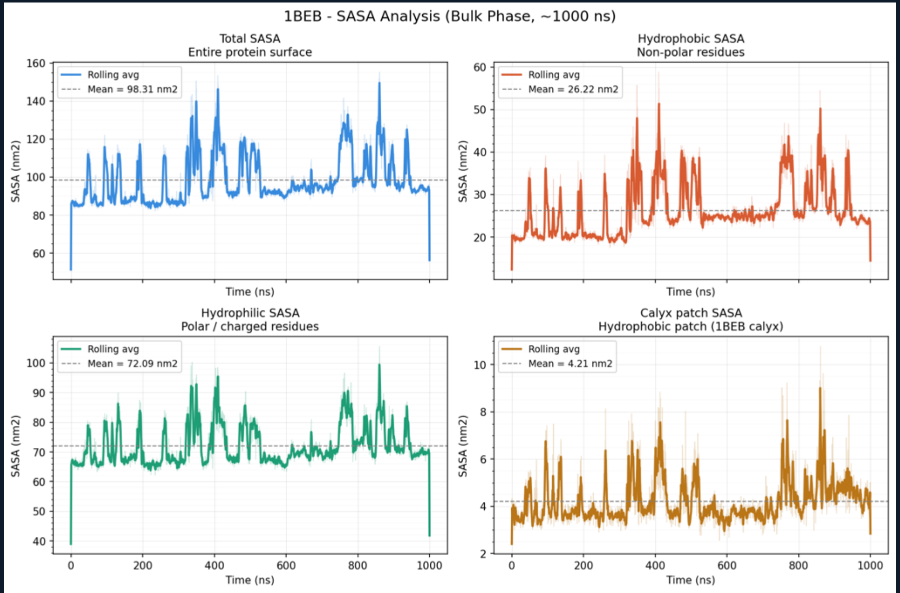


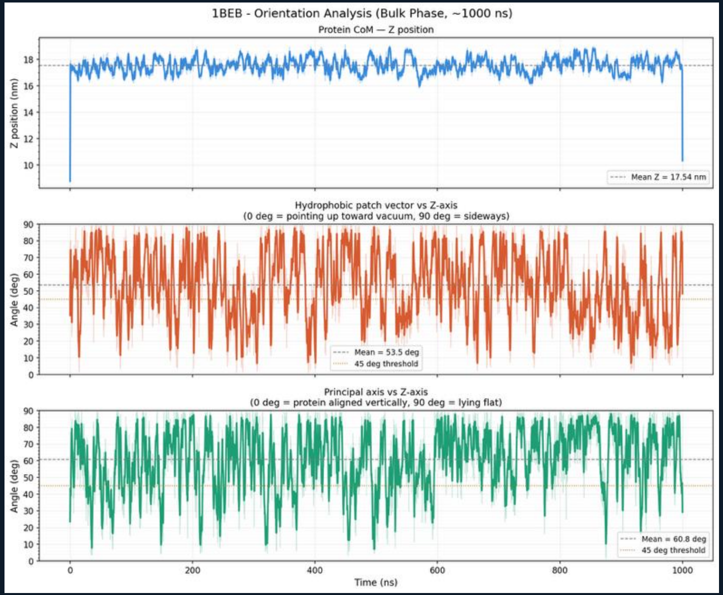


Region	Mean RMSD	RMSF note
Backbone	0.229 nm	Stable early phase
Alpha-helix	0.101 nm ✓ most stable	RMSF ~0.05–0.08 nm
Beta-sheet	0.260 nm	Disrupted after 600 ns
Loop BC (res. 30–35)	—	RMSF ~0.53 nm — most flexible

Hydrophobic Driving Force — SASA & Orientation

Increasing hydrophobic exposure over 1,000 ns





+20%
Hydrophobic SASA
0→1,000 ns

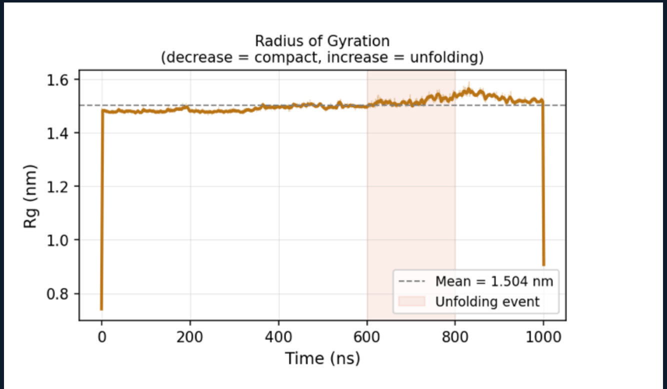
50 nm²
Peak hydrophobic
exposure (spike)

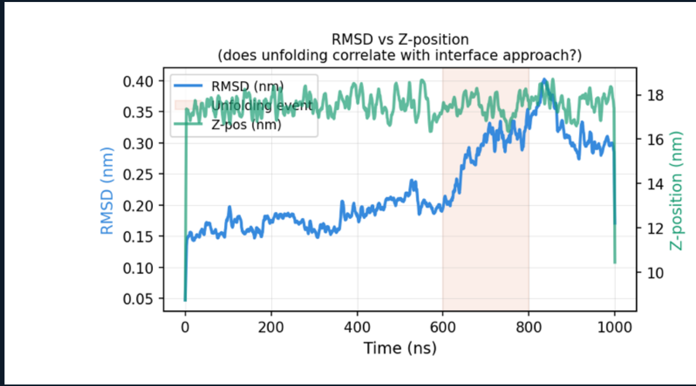
53.5°
Mean patch angle
(random tumbling)

<20°
Transient favorable
orientation events

Key Event: Partial Unfolding at 600–800 ns

Pre-adsorption conformational activation

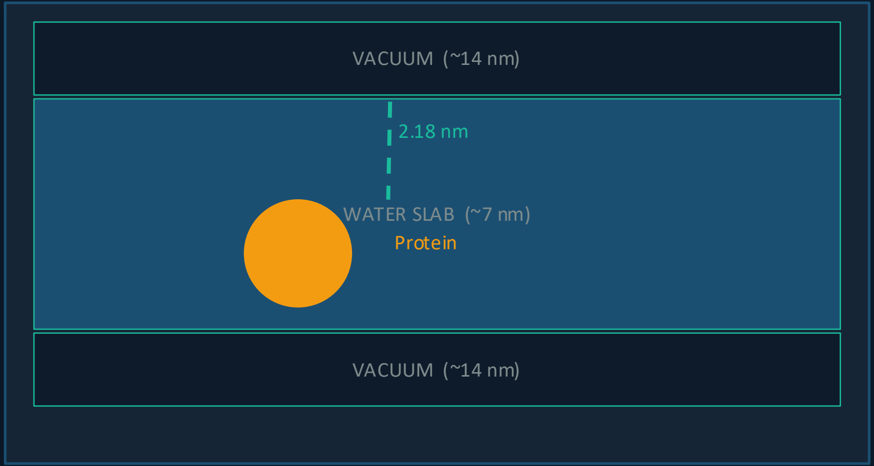




0–400 ns	400–600 ns	600–800 ns	800–1000 ns
Stable Bulk Native folded state RMSD ~0.15 nm Rg ~1.50 nm	**Pre-Activation** Conformational fluctuations increase SASA spikes more frequent	**Partial Unfolding** Rg → 1.55 nm RMSD peaks 0.4 nm Drift toward interface	**Near-Interface** Partially unfolded state persists Z-pos distribution shifts

Plan B — Near-Interface Simulation

Reducing diffusion barrier while maintaining spontaneous adsorption



Parameter	Value
Starting file	slab_planb_v2.gro
Protein CoM Z	18.700 nm
Distance to interface	2.183 nm (measured)
Still spontaneous?	YES — no orientation bias
Replicas	3 × 500 ns (seed: 1234 / 5678 / 9999)
Expected adsorption	100–300 ns

slab_planb_v2.gro → Energy Min. → NVT Slab 10 ns → Production 500 ns × 3

Summary & Research Roadmap

What we found — what comes next

Key Findings (1,000 ns)

1. Spontaneous adsorption from bulk requires >1 µs

   → first atomistic timescale quantification

2. Partial unfolding event at 600–800 ns

   → prerequisite for adsorption

3. Unfolding correlates with interface approach

   → unfolding drives adsorption tendency

4. Loop BC (res. 30–35) most flexible

   → predicted primary contact point

5. Hydrophobic SASA +20% over 1,000 ns

   → progressive pre-activation

Phase 1 — IN PROGRESS

BLG Adsorption

1,000 ns ✓ + Plan B (3 replicas)

Phase 2 — UPCOMING

Casein Adsorption

αs1-Casein (IDP) vs BLG comparison

Phase 3 — PLANNED

Calcium Bridge

BLG + Casein + Ca^{2+} at interface

Phase 4 — PLANNED

Fat Interaction

Triglyceride + protein film model